Big Data Storage and Analysis
Term Project- Group 2

Project Proposal: Traffic Accident Risk Analysis

Scenario:

This project focuses on analyzing urban traffic accident risk using an end-to-end big data pipeline built with Databricks and PySpark. The dataset simulates realistic traffic conditions to understand how environmental and human factors contribute to accidents.

Tables and Relationships:

Accidents, which records each event along with its severity and key identifiers; Locations, which captures the city, road type, and speed limit; Weather, which describes environmental conditions; Vehicles, which stores driver demographics and vehicle information; and Violations, which logs traffic offences associated with accidents.

Accidents (accident_id, location_id, weather_id, vehicle_id, severity),

Locations (location_id, city, road_type, speed_limit),

Weather (weather_id, condition, visibility, temperature),

Vehicles (vehicle_id, vehicle_type, driver_age, driver_gender),

Violations (violation_id, accident_id, violation_type).

Relationships are defined using foreign keys, such as Accidents.location_id linking to Locations.location_id, Accidents.weather_id to Weather.weather_id, and Violations.accident_id to Accidents.accident_id.

Expected outcome:

The project aims to generate insights such as identifying accident hotspots, analyzing the impact of weather and violations on severity, and enabling data-driven decision-making to improve urban road safety.